Genetic algorithms

A form of computer science algorithm where the point of the method is to cull the weak and breed the strong candidates. In other words, this is natural selection on a digital scale.

The real world has biology with 2 distinct factors that distinguish what makes a living organism unique. The Genotype and the Phenotype, mainly the genotype is the code of the organism while the phenotype is the observable difference of those traits. Both shaped by their environments and parents. Now applying this philosophy to a little ai that is tasked with something, for this example flipping a string of binary. 000000 to 111111.

First, you cant start a generation with only one instance, so like how God did it, we create two, or even more than that, then take the top performing ai and then make a new bot with the genetic code of the two, then repeat. This is the general gist of what a genetic algorithm is, now as I am currently a student in physics and I want to show off my skills, I want to make a project using this, it's also a good starting point to understanding neural networks.

EvoXRB: Genetic-Algorithm Reconstruction of a Black-Hole X-Ray Outburst

Scientific question: Can a genetic algorithm recover the physical parameters of an accreting black-hole system from real X-ray observations, and track how those parameters change as the system transitions between accretion states?

The test subject here is going to be a black-hole-X-ray-binary, **MAXI J1820+070**. It is great for this case study because NICER observed its 2018 outburst extensively, including its spectral changes. NASA reports that NICER followed the source over roughly 250 days, with hundreds of kilo seconds of exposure. (NASA, 2018) (NASA, 2018)

What I am trying to measure here

When you observe an X-ray spectrum $C(E)$, where C is the number of detected photons in different energy channels. The underlying emission from the black-hole system contains several physical components, model it as

$$F(E) = absorption \times (accretion\ disk + corona)$$

Hence, I want the parameters of the GA to discover to be something like

$$\boldsymbol{\theta} = (T_{in}, N_{disk}, \Gamma_{\mathrm{in}}, N_{PL})$$

Where T_{in} is the inner accretion-disk temperature, N_{disk} is the disk normalisation, Γ is the photon index of the high-energy power law and N_{PL} is the power-law normalisation.

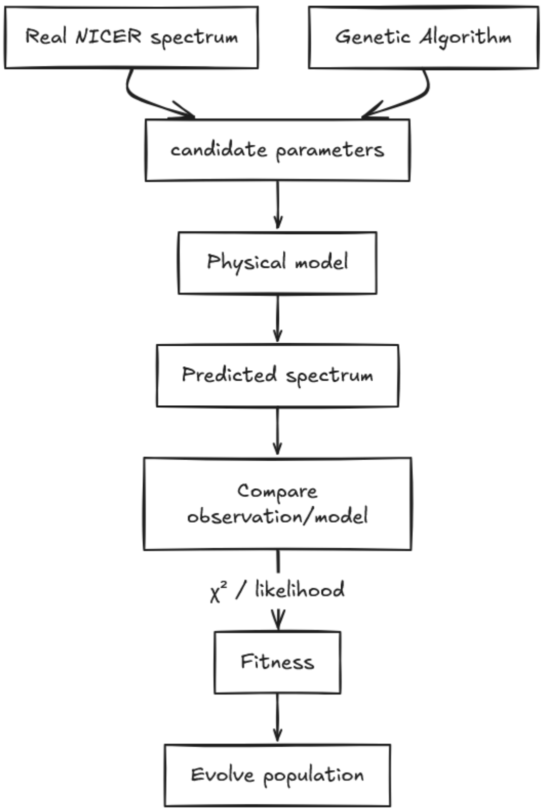

Figure 1 Conceptual understanding of the GA

The central genetic algorithm

Each chromosome represents one possible physical configuration of the source.

For instance,

T_in = 0.61 KeV
N_disk = 1830
Γ_in = 2.17
N_PL = 0.084

Could be the configuration for one individual. We then must generate the model spectra for both.

For a high-count implementation, the loss is going to initially be,

$$\chi^2 = \sum_i \frac{(C_{i.obs} - C_{i,model})^2}{\sigma_i^2}$$

The GA therefore tries to find

$$\boldsymbol{\theta}_{best} = arg \min_{\boldsymbol{\theta}} \chi^2$$

And because GAs normally maximises fitness we can define, $F = -\chi^2$, which can be eventually replaced by a Poisson statistic.

Synthetic observations

To ensure a strong methodology, the first stage of the observation is going to be purely synthetic, so we will initially define the accretion disk temperature as 0.70 keV, and the photon index as 2.10. Give these details to the genetic algorithm to generate an X-ray spectrum we should get some result that can be used to optimise and recover known physical parameters.

XSPEC provides spectral simulation capabilities, so you can generate realistic synthetic observations that include detector responses rather than simply adding random Gaussian noise to a mathematical curve.

Real observational data

As mentioned at the start, the primary dataset here is the NICER observations of **MAXI J1820+070**. (NASA, 2018)

HEASARC is specifically NASA's multi-mission archive for X-ray, gamma-ray and other high-energy astrophysics data. It includes NICER, NuSTAR, Swift, MAXI, Chandra, XMM-Newton, XRISM and numerous other missions.

So, what data do we need?

Using python and more specifically astroquery, we can gather data from the archive and the main data that is needed to be gathered is as follows,

obsid,
date,
exposure_s,
ra_deg,
dec_deg,
data_url,
state,
soft_counts,
hard_counts,
hardness_ratio,
count_rate,
Tin_keV,
Tin_error,
Gamma,
Gamma_error,
disk_norm,
powerlaw_norm,
chi_squared,
ga_generations,
ga_population

Dumping it all on a csv, then working with the observations.

Data reduction

Raw astronomical data cannot simply be thrown into NumPy and shaken until the answer comes out, although that would be great…

NICER has a reduction pipeline and the NASA workflow is as followed,

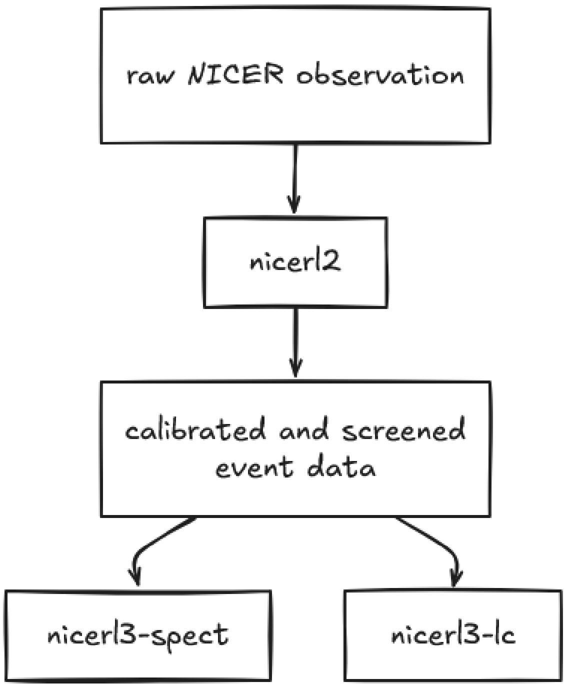

Figure 2 NASA data reduction pipeline for NICER observations

There are two results, nicerl3-spect creates the spectrum, background estimate and response files needed for spectral analysis; nicerl3-lc creates light curves. (NASA, 2023)

(As of August 2026, the current HEASoft release is **6.37**, which contains the relevant NICER tools and XSPEC. (NASA, 2026))

Instrument response

This is where the real observational astronomy comes in, the telescope does not simply measure, $F(E)$, but $F(E) \rightarrow telescope \rightarrow detector \rightarrow C_i$.

The instrument response transforms the physical photon spectrum into the counts recorded by the detector. Therefore, we need two types of files to tell me how the telescope/detector turns incoming X-ray photons into the counts that are being measured.

1. ARF or Ancillary Response File – How likely is the telescope to collect a photon of this energy

$$detected\ photons \propto F(E)Aeff(E)$$

2. RMF or Redistribution Matrix File – A detector cannot measure photon energies perfectly, the RMF gives the probability that a photon with true energy E gets recorded in a particular detector channel

$$R(i, E) = P(detected\ in\ channel\ i \mid true\ energy\ E)$$

Thankfully, NICER's nicerl3-spect can generate both automatically using the calibration database. The GA should compare the observation against a model **folded through the detector response**, not against the raw theoretical function.

Phase 1: Building the normal spectral analysis

Before implementing the genetic algorithm, we will fit one spectrum normally using XSPEC (notation of compression a physics model into an X-ray spectrum, so using something like

$$TBabs * (diskbb + powerlaw)$$

And recording,

$$T_{in}, N_{disk}, \Gamma_{\mathrm{in}}, N_{PL}$$

Which gives the baseline.

Bibliography

NASA. (2018). Retrieved from https://heasarc.gsfc.nasa.gov/docs/nicer/proposals/science_team_investigations/

NASA. (2018). Retrieved from https://heasarc.gsfc.nasa.gov/

NASA. (2018). Retrieved from https://heasarc.gsfc.nasa.gov/docs/nicer/science_nuggets/20181005.html

NASA. (2023). Retrieved from https://heasarc.gsfc.nasa.gov/docs/nicer/analysis_threads/nicerl3-spect/

NASA. (2026). Retrieved from https://heasarc.gsfc.nasa.gov/docs/software/lheasoft/download.html